V1 Matrix Core: Event Sparsity × RIA 8:16 Weight Sparsity

PS DDR / CMA

- Output Y — INT32 accumulators
- Packed RIA Weights — INT8 values + selection bytes
- Input X — INT16 activation rows

HP0 read

RIA 8:16 Weight Path

One 16-weight group: exactly 8 retained

w0	×	w2	×	w4	w5	×	×	w8	×	w10	×	w12	×	w14	×

offline packing

Hardware Tile Format

Weight Tile RAM	8 × INT8 values
Selected Tile RAM	8-bit lane mask

input-feature-major, 8 output lanes / tile

w + selected[7:0]

Weight / Mask Selection
mask[i][lane] = 1 ?

Input Event Path

ExSpike RTL adapted to 16-element groups

Input Tile RAM
64-bit AXI beats

$x_i \neq 0$

16-bit Non-zero Event Map

1	0	1	0	0	1	…	0

FAST_FILTER
lowest-set-bit iterator

one-hot

INDEX2ABS_SUB_0
one-hot → absolute index

Valid Event
index i
signed activation $x_i \neq 0$

activation value

AXI-Lite Configuration + FSM
vectors · in/out features · addresses · start

HP0 write

lane mask + weights

index + value

Event-driven Sparse Compute

event ∩
RIA mask

selected lanes only

8-lane Sparse MAC
INT16 × INT8

partial products

8 × MP_ACC
48-bit signed accumulation

Saturate to INT32
AXI write-back

instrumentation

Hardware Counters

cycles · events · selected MAC issues · read beats

Two independent sparsity conditions

Event sparsity: $x_i = 0$ → no event, no MAC issue

Weight sparsity: exactly 8 of every 16 weights retained

Executed work: valid input event ∩ retained-weight mask

Event / tensor flow · Weight / mask flow · Signed activation value

Sparse arithmetic · Control / counters